Data Science Capstone – Lightning Talk 1 (The Pitch)

The **Lightning Talk** is your first major milestone in the Capstone project.
You will have **5-7 minutes** to convince your audience that you have identified a **real, valuable problems (5 problems at least) worth solving with data**.

This talk is **not about showing technical depth**. It is about **selling the problem, the idea, and the feasibility of your solution**—quickly and clearly.

Objective of the Lightning Talk

By the end of your presentation, your peers and instructors should clearly understand:

- The **real-world problem** you want to solve
- Why this problem **matters**
- How **Data Science / Machine Learning** can help
- What **data** you will use
- How you will measure **success**

You are pitching the **value and feasibility** of your project.

Recommended Structure

Use the **Hook → Problem → Solution → Data → Success Metrics** framework.

1. The Hook

What to include

- Title
- Your name

- One strong visual, fact, or statistic

2. The Problem

What to include

- What pain point are you solving?
- Who is affected (Potential Audience)?
- Why does this matter?

3. Proposed Solution

What to include

- How can Data Science or ML help?
- High-level approach only (e.g., classification, regression)

4. The Data Source(s)

What to include

- Where the data comes from (CSV, API, database, website)
- Key features or variables

5. Success Metrics

What to include

- How will you know your model works? What does success look like?
- Example metrics: Accuracy, RMSE, Precision, Recall

How to Find a Real Problem

1. Look for the Gap

- Inefficiency: “This takes 20 hours—can it be automated?”
- Risk: “Failures happen suddenly—can we predict them?”

2. Think Like a Job Candidate

Pick problems in the industry you want to work in:

- Healthcare: readmissions, diagnosis, medical images
- E-commerce: churn, recommendations, sentiment

This makes your portfolio more attractive to recruiters.

3. Use Pain-Point Questions

Ask yourself:

- What is expensive/slow/annoying?
- What insight do I wish existed?

Where to Find Proven Problems

1. Job Boards (Upwork / Freelancer)

Search for ML, NLP, CV, LLM projects.
If someone is paying for it, it’s a **real problem**.

2. Startup Failures & Hackathons

- Failed startups often reveal unsolved data problems
- Product Hunt winners validate real demand
- Your project can be a **smaller MVP**

3. Reddit = Pain-Point Goldmine

Useful subreddits:

- r/smallbusiness
- r/startups

Turning a Found Problem into a Pitch

Use this template:

- **Evidence**: “Many small business owners struggle with X”
- **Current solution**: “They do this manually in Excel”
- **Your MVP**: “I will build a model that automates this using Y”

Final Thought: The Clone Strategy

Take one feature from a successful data-driven product and adapt it:

- “Zillow for used cars”
- “Duolingo for learning to code”

Small, focused, and achievable wins every time.

Common Pitfalls to Avoid

- **Tools comes first:** The tool is secondary. The problem is primary.
- **Scope creep**: Don’t try to solve everything
- **No confirmed data**: Great idea, no dataset = no project
- **Too much jargon**: Keep business logic clear and simple

Useful Resources

- How to give a good lightning talk.